

COL362/632/7362 Introduction to Database Management Systems

Functional Dependencies

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Example

- ▶ Consider the following relation

ActorFilm						
id	firstname	lastname	dob	title	year	language
1	Priyanka	Chopra	1992	Don	2006	Hindi
1	Priyanka	Chopra	1992	Don-II	2011	Hindi
2	Anthony	Hopkins	1937	MI-IV	2011	English
2	Anthony	Hopkins	1937	Valkyrie	2008	English
3	Bill	Nighty	1949	Valkyrie	2008	English

- ▶ Problems with this design

1. Redundancy
2. Update Anomalies

- ▶ `update ActorFilm set firstname = 'Priyankaa' where id = 1;`
- ▶ `insert into ActorFilm values (null, null, null, null, Avatar, 2009, English);`
- ▶ `delete from ActorFilm where title = 'Valkyrie';`
- ▶ `delete from ActorFilm where firstname = 'Anthony' and lastname = 'Hopkins';`

Normalization

The process of systematically eliminating redundancy and anomalies.

Functional Dependencies

- ▶ A **functional dependency** is another kind of constraint
- ▶ If two tuples in a relation agree on the values of one set of attributes, then they must also agree on the values of another set of attributes

Given two attribute sets A and B with $A, B \subseteq [R]$

$A \rightarrow B$ is called functional dependency (FD) if a function exists $f(a) := b \quad \forall a \in \pi_A(R), b \in \pi_B(R)$ for all possible instances of R .

- ▶ In other words, values of A determine the values of B

Functional Dependencies

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Functional Dependencies (FDs)

- $\{ \text{id, firstname, lastname} \} \rightarrow \{ \text{dob} \}$
- $\{ \text{title} \} \rightarrow \{ \text{language} \}$
- $\{ \text{firstname, lastname} \} \rightarrow \{ \text{title, year} \}$

Incorrect FD!

Functional Dependencies

- ▶ Important to differentiate between instance and schema
- ▶ FD defines a constraint for the possible instances of R
 - Instances that satisfy FD are called **legal instances**
- ▶ FD do not define dependencies for concrete instance
 - Single counter example is enough!

Functional Dependencies

Example

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Functional Dependencies

- $\{ \text{id} \} \rightarrow \{ \text{firstname}, \text{lastname}, \text{dob} \}$
- $\{ \text{firstname} \} \rightarrow \{ \text{lastname} \}$ **not an FD**
- $\{ \text{dob} \} \rightarrow \{ \text{id}, \text{firstname}, \text{lastname} \}$ **not an FD**

Functional Dependencies

- ▶ Not just a theoretical DB concept!
- ▶ Play a crucial role in applications beyond DB design
 - **Feature engineering and selection in ML** (e.g., dimensionality reduction, feature generation)
 - **Data cleaning and anomaly detection** (e.g., missing value imputation, duplicate detection, data errors)
 - **Causal discovery and data understanding in ML**
 - **Query Optimization in Data pipelines**
 - **Data integration and schema matching**
 - **Balancing data privacy and data utility**

Published in ACM SIGMOD 2026!

ÆGIS: A Correlation-Based Data Masking Advisor for Data-Sharing Ecosystems

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Data-sharing ecosystems connect providers, consumers, and intermediaries to facilitate the exchange and use of data for a wide range of downstream tasks. In sensitive domains such as healthcare, privacy is enforced as a hard constraint—any shared data must satisfy a minimum privacy threshold. However, among all masking configurations that meet this requirement, the utility of the masked data can vary significantly, posing a key challenge: how to efficiently select the *optimal* configuration that preserves maximum utility. This paper presents ÆGIS, a middleware framework that selects optimal masking configurations for machine learning datasets with features and class labels. ÆGIS incorporates a utility optimizer that minimizes *predictive utility deviation*—quantifying shifts in feature-label correlations due to masking. Our framework leverages limited data summaries (such as 1D histograms) or none to estimate the feature-label joint distribution, making it suitable for scenarios where raw data is inaccessible due to privacy restrictions. To achieve this, we propose a joint distribution estimator based on iterative proportional fitting, which allows supporting various feature-label correlation quantification methods such as mutual information, chi-square, or g3. Our experimental evaluation of real-world datasets shows that ÆGIS identifies optimal masking configurations over an order of magnitude faster, while the resulting masked datasets achieve predictive performance on downstream ML tasks on par with baseline approaches and complements privacy anonymization data masking techniques.

CCS Concepts: • **Information systems** → **Data management systems**; • **Security and privacy** → **Data anonymization and sanitization**.

Additional Key Words and Phrases: Data Markets, Data Sharing Ecosystems, Data Masking, Compliant Data Management

ACM Reference Format:

Omar Islam Laskar, Fatemeh Ramezani Khozestani, Ishika Nankani, Sohrab Namazi Nia, Senjuti Basu Roy, and Kaustubh Beedkar. 2025. ÆGIS: A Correlation-Based Data Masking Advisor for Data-Sharing Ecosystems. *Proc. ACM Manag. Data* 3, 6 (SIGMOD), Article 292 (December 2025), 26 pages. <https://doi.org/10.1145/3769757>

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Super Keys

- ▶ A **superkey** is a set of one or more attributes (columns) in a relation (table) such that the values of these attributes are sufficient to uniquely identify a tuple (row) in the relation.
- ▶ Any subset $S \subseteq [R]$ for which $S \rightarrow [R]$ holds is called a **super key** of R
 - $[R] \rightarrow [R]$
 - $\exists a \in S : S \setminus \{a\} \rightarrow [R]$
- ▶ Super key functionally determines all other attributes
 - Not necessarily minimal!
- ▶ **Example**
 - ActorFilm(id, firstname, lastname, dob, title, year, language)
 - $S = \{ \text{id, firstname, lastname, dob, title} \}$

Candidate Key

- ▶ A **candidate key** is a **minimal superkey**
- ▶ For a superkey $S \subseteq [R]$ for which
 - $\nexists a \in S : S \setminus \{a\} \rightarrow [R]$ or
 - $\forall a \in S : S \setminus \{a\} \nrightarrow [R]$
 - is called the **candidate key** of R
- ▶ **Example**
 - ActorFilm(id, firstname, lastname, dob, title, year, language)
 - $S = \{ \text{firstname, lastname, title, year} \}$
- ▶ A relation can have **multiple candidate keys**

Primary Attribute & Primary Key

- ▶ Let C be the set of all candidate keys of $[R]$, then
- ▶ all attributes $\bigcup_i c_i$ are called **primary attributes** of $[R]$

Primary Key

- ▶ Any candidate key used to identify a tuple!
 - Cannot have null
 - Not change over time
 - Short as possible

Inferring FDs

- ▶ Given a set of FDs, which other FDs follow from it?
- ▶ Example
 - Given: $R(A, B, C)$ satisfies $A \rightarrow B$ and $B \rightarrow C$
 - Inferred: $A \rightarrow C$ through **transitivity of FDs**.
 - Given: $\{\text{aadhar}\} \rightarrow \{\text{pan}\}, \{\text{pan}\} \rightarrow \{\text{dob}\}$
 - Then, $\{\text{aadhar}\} \rightarrow \{\text{dob}\}$

Inferring FDs

- ▶ Given: $A_1A_2 \dots A_n \rightarrow B_1B_2 \dots B_m$
- ▶ Inferred $A_1A_2 \dots A_n \rightarrow B_i$ for $i = 1, 2, \dots, m$ through **splitting rule**

- ▶ Given $A_1A_2 \dots A_n \rightarrow B_i$ for $i = 1, 2, \dots, m$
- ▶ Inferred: $A_1A_2 \dots A_n \rightarrow B_1B_2 \dots B_m$ through **combining rule**

Equivalent FDs

- ▶ Two FDs S and T are equivalent if
 - set of relation instances satisfying S is exactly the same as the set of relation instances satisfying T
- ▶ If FD S follows from FD T , if every instance that satisfies T also satisfies S
- ▶ $S = T$ iff $S \implies T$ and $T \implies S$

Trivial and Non-trivial FDs

- ▶ Consider $R(A, B, C)$
- ▶ Given FDs
 - $AB \rightarrow A$ trivial
 - $AB \rightarrow AC$ non-trivial
 - $AB \rightarrow C$ completely non-trivial
- ▶ **Trivial FD** $X \rightarrow Y$ where $Y \subseteq X$
- ▶ **Non-trivial FD** $X \rightarrow Y$ where $Y \not\subseteq X$
- ▶ **Completely non-trivial FD** $X \rightarrow Y$ where $Y \cap X = \emptyset$

Armstrong's Axioms

- | | | | | |
|----|-------------------------------------|------------|------------------------------------|----------------------------------|
| 1. | $B \subseteq A$ | $\implies$ | $A \rightarrow B$ | Reflexivity |
| 2. | $A \rightarrow B$ | $\implies$ | $AC \rightarrow BC$ | Augmentation |
| 3. | $A \rightarrow B, B \rightarrow C$ | $\implies$ | $A \rightarrow C$ | Transitivity |
| 4. | $A \rightarrow BC$ | $\implies$ | $A \rightarrow B, A \rightarrow C$ | Decomposition (splitting) |
| 5. | $A \rightarrow B, A \rightarrow C$ | $\implies$ | $A \rightarrow BC$ | Union (combining) |
| 6. | $A \rightarrow B, DB \rightarrow C$ | $\implies$ | $DA \rightarrow C$ | Pseudo-transitivity |

Closure of FDs

- ▶ Given: the set S of FDs
- ▶ Output: S^+ , the **closure** of S containing all FDs that can be derived from S
- ▶ Example
 - Given $S = \{ AB \rightarrow C, C \rightarrow ED \}$
 - Then $S^+ = \{ AB \rightarrow ED, AB \rightarrow E, AB \rightarrow D, ABC \rightarrow ED, C \rightarrow E, C \rightarrow D, \dots \}$
 - HW: Find all FDs

Closure of Attributes

- ▶ Let S be a set of FDs
 - Is a new FD F derivable from S ?
- ▶ Example
 - Given
$$S = \{ AB \rightarrow C, BC \rightarrow AD, D \rightarrow E, CF \rightarrow B \}$$
 - Does the FD $AB \rightarrow D$ hold?
 - Compute $\{A, B\}^+$

FD	$\{A, B\}^+$
$AB \rightarrow C$	$\{A, B, C\}$
$BC \rightarrow AD$	$\{A, B, C, D\}$
$D \rightarrow E$	$\{A, B, C, D, E\}$
 - $AB \rightarrow D$ holds!

Attribute Closure Method

Given

- ▶ set S of FDs
- ▶ candidate FD $F : A \rightarrow B$

Step 1: Compute A^+

$A^+ = A$ initial set of attributes

```
do {  
  for each FD  $X \rightarrow Y \in S$  {  
    if  $X \subseteq A^+$  then  $A^+ \leftarrow A^+ \cup Y$   
  }  
  if  $A^+$  unchanged, quit  
}
```

Step 2: Check if $B \subseteq A^+$